Correction to The Coherence-Gated Yang-Mills Mass Gap

A Finite-Volume Gauge-Scalar Model and Open Quantum Limits

Author: Micah Blumberg

Self Aware Networks Research Institute

https://selfawarenetworks.com

July 27, 2026

Abstract

Version 1 claimed a positive infinite-volume Yang-Mills mass gap from a coherence-gated gauge-scalar model. The displayed estimate was

```
Delta*(L) >= min{Delta_coh, c_P L^-1 sqrt(kappa_F)},
c_P = 2*pi/L.                                                        (1)
```

Its gauge term is therefore `2*pi*sqrt(kappa_F)/L^2` and tends to zero. The minimum lower bound also tends to zero, so the published thermodynamic-limit conclusion does not follow. This correction retracts the infinite-volume spectral-gap and Millennium-problem claims. It retains a narrower object: a classical gauge-scalar energy on a fixed three-torus and an exact Poincare coercivity estimate for its Abelian linearized quadratic form after transverse, mean-zero, gauge, and harmonic-mode projection. Quantization, construction of the physical Hilbert space, gauge reduction, self-adjointness, continuum and thermodynamic limits, and any volume-uniform gap remain open. An explicit first-Fourier-mode witness also shows that the retained vector coercivity coefficient is sharp in this projected quadratic model and collapses as the box grows; no volume-uniform constant can be recovered from that quadratic form by improving the estimate. Because the model adds scalar fields and scalar-dependent gauge dressing, it is not the pure four-dimensional Yang-Mills theory specified by the Clay problem [1]. The revised contribution is a transparent finite-volume model and a precise list of the mathematical obligations required for further progress. It reports no new positive result for pure Yang-Mills theory.

Scope and evidence boundary

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Continues: doi:10.5281/zenodo.21463268
Concept DOI: doi:10.5281/zenodo.17311156

Reader Orientation: The Problem Before the Terminology

The central mistake can be understood without specialized language. A finite periodic box has a longest allowed wavelength and therefore a smallest nonzero Fourier frequency. That frequency is positive for every fixed box, but it moves toward zero when the box is enlarged. A positive estimate that depends on that frequency is therefore not automatically a positive estimate for an infinite system.

The correction has three steps:

1. substitute the paper's own definition into its published lower bound;
2. separate a classical energy inequality from a quantum spectral-gap claim; and
3. exhibit an allowed first Fourier mode that attains the shrinking vector coefficient.

Only after those operations are clear is the historical phrase **coherence gating** needed. In Version 1, it named the proposed idea that every physical gauge excitation must activate a positive-cost scalar sector. The paper did not construct the physical quantum state space or prove that universal activation statement. Version 3 therefore treats coherence gating as an unproved proposed mechanism, not as a synonym for the elementary finite-volume inequality.

This ordering is deliberate: ordinary problem first, mathematical distinction second, inherited terminology third, and failure conditions last.

1. Correction and Scope

The published PDF remains the historical baseline. This draft changes the central result rather than merely adjusting notation. The following claims are withdrawn:

1. the displayed bound proves a positive thermodynamic-limit gap;
2. every gauge-invariant excitation necessarily pays a fixed coherence cost;
3. the listed assumptions construct a physical quantum Hamiltonian and Hilbert space;
4. strong-resolvent language alone transfers the bound to a continuum theory;
5. the coherence-gated gauge-scalar model solves the pure Yang-Mills Millennium problem.

The retained target is deliberately smaller but substantive: specify a classical finite-volume gauge-scalar model, prove the exact coercivity estimate that follows from its quadratic form, identify the zero modes and assumptions, prove the coefficient's sharpness inside that quadratic problem, and state what fails to survive as volume grows.

Equation (1) is reproduced solely to audit the Version 1 inference. Version 3 does not assert that the formula has a validated physical derivation or a complete dimensional interpretation; the decisive correction is the algebraic substitution within the formula as it was published.

Version 1 statement	Version 3 disposition	Bounded replacement
Positive thermodynamic-limit gap	Retracted	The displayed lower bound tends to zero
Universal coherence activation	Open and unproved	Exact activation theorem would require a constructed physical state space
Quantum Hamiltonian and Hilbert space established	Retracted	Regulator, reduction, domains, and limits are explicit obligations
Pure Yang-Mills Millennium problem solved	Retracted	The paper studies a different gauge-scalar model
Finite-volume Poincare estimate	Retained with boundary	Proposition 1 is classical, Abelian, linearized, projected, and fixed-volume

2. The Model Is Gauge-Scalar, Not Pure Yang-Mills

Let the spatial domain be the flat torus `T_L^3`. Consider a gauge connection `A`, a complex scalar `psi`, and any additional scalar background `tau`. A schematic classical energy is

```
E_L[A,psi,tau] = integral_(T_L^3) [
    kappa_F f(tau,|psi|) |F_A|^2/4
  + kappa_psi |D_A psi|^2
  + kappa_tau |grad tau|^2/2
  + V(tau,|psi|)
] dx.                                                    (2)
```

This definition requires a compact gauge group, representation, coupling, regularity class, gauge quotient, and boundary conditions. Those data must be fixed in any theorem. Non-Abelian gauge fixing also encounters global orbit obstructions documented by Gribov and Singer [5,6]. The scalar and dressing function make the model a gauge-scalar theory in the comparison domain of gauge-Higgs systems [2]. Pure Yang-Mills is not recovered merely by calling the scalar a coherence field.

The function `f` must be stated exactly. The inequality

```
f(tau,|psi|) >= c_f |psi|^2.                                        (3)
```

does not imply `f(tau,0)=0`. If vanishing at zero is desired, it is a separate assumption. A vanishing gauge kinetic coefficient may also make the principal part degenerate; it is not equivalent to a gauge-boson mass and requires its own well-posedness analysis.

3. A Valid Fixed-Volume Coercivity Proposition

The cleanest surviving statement concerns a specified Abelian linearized quadratic form, not the full nonlinear energy in Section 2. Let

```
T_L^3 = (R/LZ)^3,     k_min=2*pi/L.                                 (4)
```

Use the ordinary unnormalized `L2(T_L^3)` norm. Let `u` be a real mean-zero field in `H^1(T_L^3)`. Let `a` be a real vector field in `H^1(T_L^3;R^3)` satisfying

```
div a=0,     integral_(T_L^3) a dx=0.                               (5)
```

The transverse condition removes longitudinal gauge directions in this linearized Abelian sector. The zero-mean condition removes the constant harmonic one-forms. It does not classify non-Abelian gauge copies, topological sectors, or nonlinear constraints.

For `kappa_u>0`, `kappa_F>0`, and `m_u>=0`, define

```
Q_L[u,a] = kappa_u ||grad u||_2^2
         + m_u^2 ||u||_2^2
         + kappa_F ||curl a||_2^2.                                  (6)
```

Here `u` and `a` are independent projected fields in the stated Abelian quadratic problem. Proposition 1 does not derive them by linearizing Equation (2), and it does not transfer the result to that schematic nonlinear energy.

Proposition 1 (fixed-volume projected quadratic coercivity). Under the displayed assumptions,

```
Q_L[u,a] >= (m_u^2+kappa_u k_min^2)||u||_2^2
              + kappa_F k_min^2||a||_2^2.                           (7)
```

Consequently, for the direct-sum norm `||(u,a)||^2=||u||_2^2+||a||_2^2`,

```
Q_L[u,a] >= min{m_u^2 + kappa_u k_min^2,
                kappa_F k_min^2}
               (||u||_2^2 + ||a||_2^2).                             (8)
```

Proof. Expand both fields in Fourier modes `k=2*pi*n/L`, `n in Z^3`. The zero-mean hypotheses remove `n=0`. Parseval's identity then gives

```
||grad u||_2^2 = sum_(k != 0) |k|^2 |u_hat(k)|^2
               >= k_min^2 ||u||_2^2.                                (9)
```

For every transverse vector coefficient, k \dot\a_hat(k)=0, so `|k cross a_hat(k)|^2=|k|^2|a_hat(k)|^2`. Hence

```
||curl a||_2^2 >= k_min^2 ||a||_2^2.                                    (10)
```

Multiplying by the positive coefficients and adding the mass term proves the componentwise inequality. Taking the smaller coefficient proves the direct-sum corollary. No quantization or spectral theorem enters this proof.

This is a standard direct Fourier / Parseval estimate. Its role in Version 3 is to identify exactly what survives the correction, not to claim a novel Yang-Mills theorem.

The direct-sum coefficient is positive for each fixed finite `L`. The vector coefficient is `kappa_F(2*pi/L)^2`, however, so it scales as `L^-2` and the minimum tends to zero as `L` tends to infinity. The scalar mass term does not prevent this because the direct sum includes admissible projected vector perturbations with `u=0`.

Proposition 2 (sharpness and failure of volume-uniform coercivity in the projected quadratic model). The two component constants in Equation (7) are attained by first nonzero Fourier modes. In particular, take a_L(x)=e_2\cos(2*pi*x_1/L) and `u=0`. Then `a_L` is real, transverse, and mean-zero, and

```
Q_L[0,a_L] / ||a_L||_2^2
    = kappa_F (2*pi/L)^2 -> 0 as L -> infinity.                         (11)
```

Likewise, u_L(x)=cos(2*pi*x_1/L) with `a=0` attains the scalar coefficient `m_u^2+kappa_u(2*pi/L)^2`. Thus the constants in Equation (7), and the smaller constant in Equation (8), are optimal for the stated quadratic form.

Proof. Both fields contain only the two Fourier modes with wavevectors `+/- (2*pi/L)e_1`. For the vector witness, `e_1 dot e_2=0`, so the mode is transverse and the curl identity used in Equation (10) is an equality. For the scalar witness, the Poincare inequality in Equation (9) is an equality. Substitution into Equation (6) gives the displayed ratios.

Proposition 2 is stronger than merely observing that one lower bound happens to vanish. It proves that no larger volume-uniform coercivity constant exists for `Q_L` on the stated direct-sum space, because an admissible normalized sequence has energy ratio tending to zero. It still says nothing by itself about the spectrum of a different nonlinear, non-Abelian, constrained quantum theory.

The two propositions do not establish:

- a unique classical vacuum across all topological sectors;
- a self-adjoint quantum Hamiltonian;
- a gap between quantum energy eigenvalues;
- a gauge-invariant continuum measure;
- confinement or an area law; or
- a positive volume-uniform lower bound.

It also does not prove coercivity of the nonlinear non-Abelian energy written schematically in Section 2. Any such extension must specify a gauge slice, control nonlinear terms and gauge copies, handle topology and degeneracy of `f`, and prove constants in the appropriate function spaces.

4. Why the Published Limit Fails

The earlier estimate used both `c_P=2*pi/L` and another factor `L^-1`:

```
b_L = c_P L^-1 sqrt(kappa_F)
    = 2*pi*sqrt(kappa_F)/L^2.                                           (12)
```

Thus `b_L -> 0`. For fixed positive `Delta_coh`, there is an `L_0` after which `b_L < Delta_coh`, and then

```
min{Delta_coh,b_L}=b_L -> 0.                                    (13)
```

The inference

```
liminf Delta*(L) >= Delta_coh
```

does not follow. A lower bound tending to zero does not prove that the actual gap tends to zero; it simply gives no positive infinite-volume information. The executable check in model/check_gap_limit.py verifies this substitution, the retained componentwise coefficients, their fixed-volume positivity, and their lack of volume uniformity, including the first-mode sharpness witness.

5. The Coherence-Activation Gap

The published proof used two incompatible routes:

- a gauge-observable route said coherence activation is unavoidable; and
- a separate case allowed a pure-gauge excitation with no coherence activation.

If the latter states are physical, their only displayed floor vanishes with volume. If they are not physical, their exclusion must be proved from the exact constraints and operator algebra rather than asserted in prose.

More generally, multiplying a gauge kinetic term by a scalar function does not show that every nonvacuum gauge-invariant state contains a scalar excitation with fixed energy. A replacement theorem would need:

1. a constructed physical state space;
2. a precise algebra of gauge-invariant observables;
3. a proof that every state orthogonal to the vacuum lies in a sector with a scalar lower bound; and
4. constants uniform in regulator and volume.

No such theorem is claimed here.

6. Quantum Construction Obligations

A quantum spectral-gap statement cannot assume the main objects it is meant to construct. At minimum, future work must specify and establish:

6.1 Regulated theory

Choose a lattice or another regulator, define the gauge group and scalar representation, construct the gauge-invariant Hilbert space, define the Hamiltonian on a dense domain, and establish self-adjointness and a lower bound. Wilson's lattice formulation and the Kogut-Susskind Hamiltonian are historical construction anchors, not proofs for the present model [3,4].

6.2 Vacuum and sectors

Classify gauge, harmonic, topological, and scalar sectors. Prove uniqueness if it is claimed. State whether the gap is sector-specific or global.

6.3 Continuum limit

Provide a sequence of regulated theories and prove convergence strong enough to preserve the required axioms and observables. Osterwalder-Schrader reconstruction illustrates the strength of the conditions needed to pass from Euclidean functions to a quantum field theory [7]. Strong-resolvent convergence alone does not automatically preserve a positive gap without a uniform spectral-isolation argument; operator convergence and spectral stability are separate obligations [8].

6.4 Thermodynamic limit

Derive a lower bound whose positive constant is independent of `L`, or prove a different volume-uniform mechanism. A finite-volume Poincare constant cannot serve that role because its first nonzero mode collapses with volume.

6.5 Relation to pure Yang-Mills

Either remove the scalar and prove results for the pure theory or state a rigorous limiting map under which the added fields decouple while the desired gap and axioms persist. Without such a theorem, the gauge-scalar model remains a different theory.

7. What Would Count as Progress

The corrected research program has several meaningful intermediate targets:

1. prove existence and coercivity for the classical finite-volume energy in a named gauge and function space;
2. characterize all zero modes and topological sectors;
3. formulate a lattice Hamiltonian with a machine-checkable assumption ledger;
4. compute finite-volume spectra as exploratory evidence with pinned code and convergence tests;
5. derive, rather than assume, any scalar-induced gauge mass mechanism;
6. prove a regulator- and volume-uniform estimate if one exists; and
7. compare the resulting theory honestly with lattice gauge-Higgs literature rather than treating added scalar fields as pure Yang-Mills [2].

Finite-volume numerical spectra would be computational evidence only. They would not establish a continuum quantum field theory or the Clay target.

8. Relation to SIT and Branch C

The parent SIT action and the Branch C arithmetic program do not repair this limit. SIT Version 79 has its own correction obligations. Branch C packets and Lean artifacts establish only their exact finite claim signatures; BC-055O is a finite listed-mediator constructor result, not a Yang-Mills theorem. No cross-paper citation changes the status of the gauge-scalar model.

9. Claim and Release Discipline

This revision uses the following language:

- **proved here** only for the displayed classical fixed-volume coercivity proposition;
- **conditional** for consequences of unconstructed quantum objects;
- **computational** for future regulated numerical checks;
- **open** for continuum, thermodynamic, confinement, and mass-gap questions; and
- **not addressed** for the pure Yang-Mills Millennium problem.

Before repository release, the source must pass equation, dimension, reference, and claim-status audits; the executable scaling check; rendered-PDF review; hash generation; and DOI-lineage review. Independent gauge-theory review remains recommended but is not classified as unfinished local drafting work. These checks qualify the correction / model note for release; they do not close the nonlinear, quantum, continuum, thermodynamic, or Clay problems.

10. Conclusion

The published estimate is positive at fixed finite volume but supplies no positive infinite-volume lower bound. Correcting that point changes the paper's thesis. The defensible Version 3 result is the exact failure of the published limit, Proposition 1's finite-volume projected classical inequality, Proposition 2's sharp first-mode obstruction to volume-uniform coercivity in that quadratic model, the executable regression surface, and the explicit quantum-construction obligations. This is a legitimate mathematical-physics correction and model note, but it is neither a proof of an infinite-volume spectral gap nor a solution of the pure Yang-Mills Millennium problem.

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